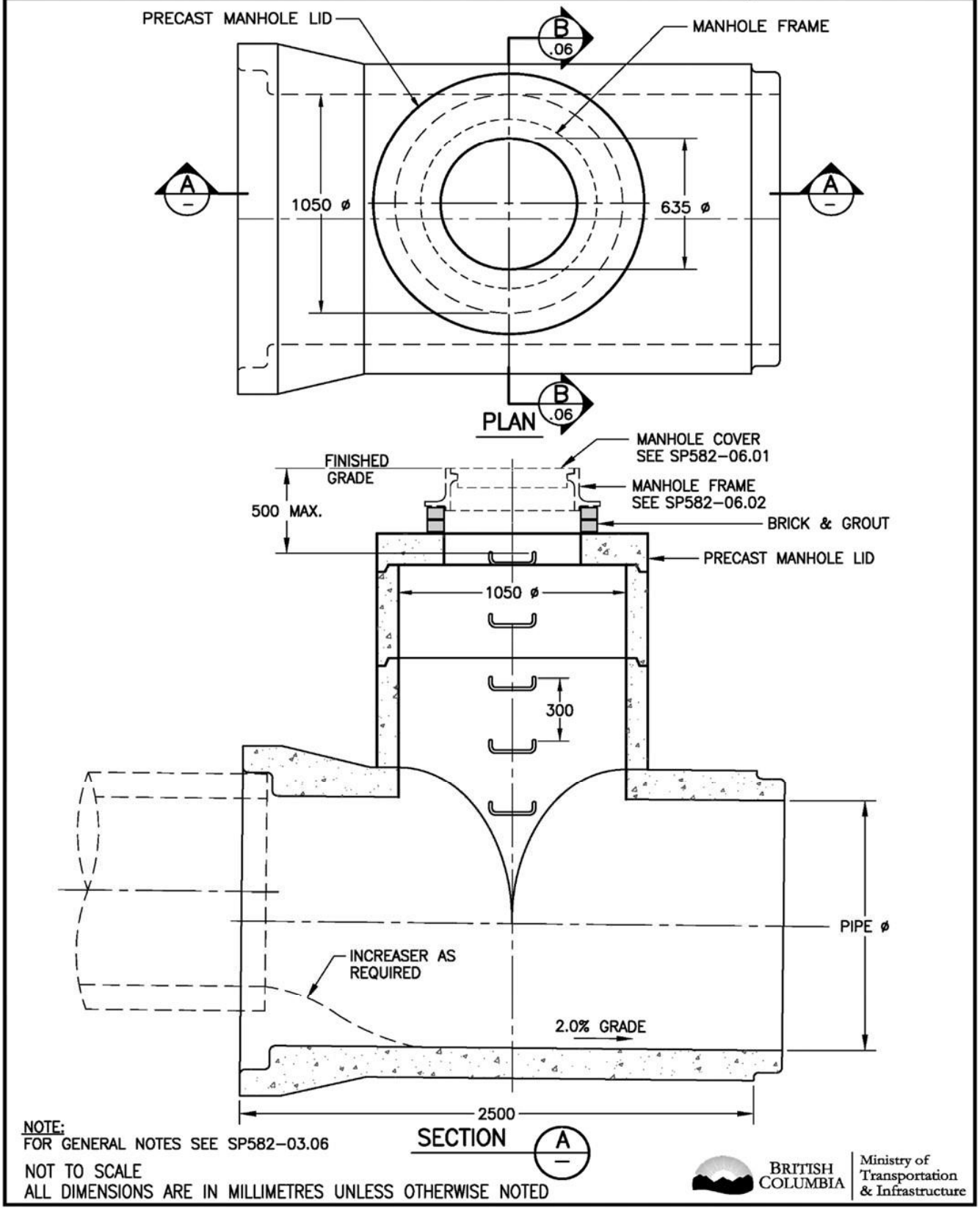
PREFABRICATED CONCRETE TEE MANHOLES PIPE SIZES – 750 mm TO 3050 mm

SP582–03.05



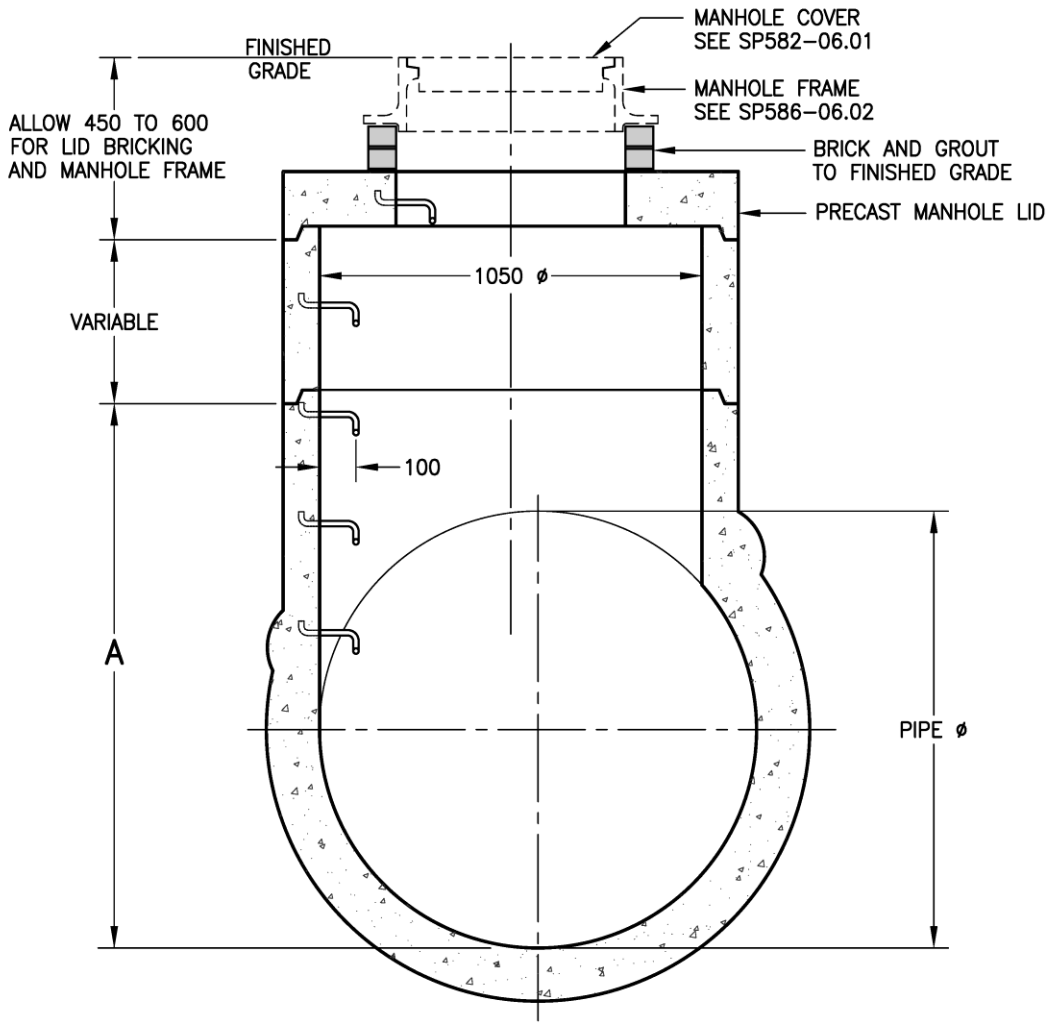
NOTE:
FOR GENERAL NOTES SEE SP582–03.06

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

PREFABRICATED CONCRETE TEE MANHOLES PIPE SIZES – 750 mm TO 3050

SP582–03.06



SECTION B .05

NOTE: FOR LOCATION OF SECTION SEE SP582–03.05

PIPE ø	A
750	1270
900	1400
1050	1550
1200	1720
1350	1880
1500	1970
1800	2210

NOTES:

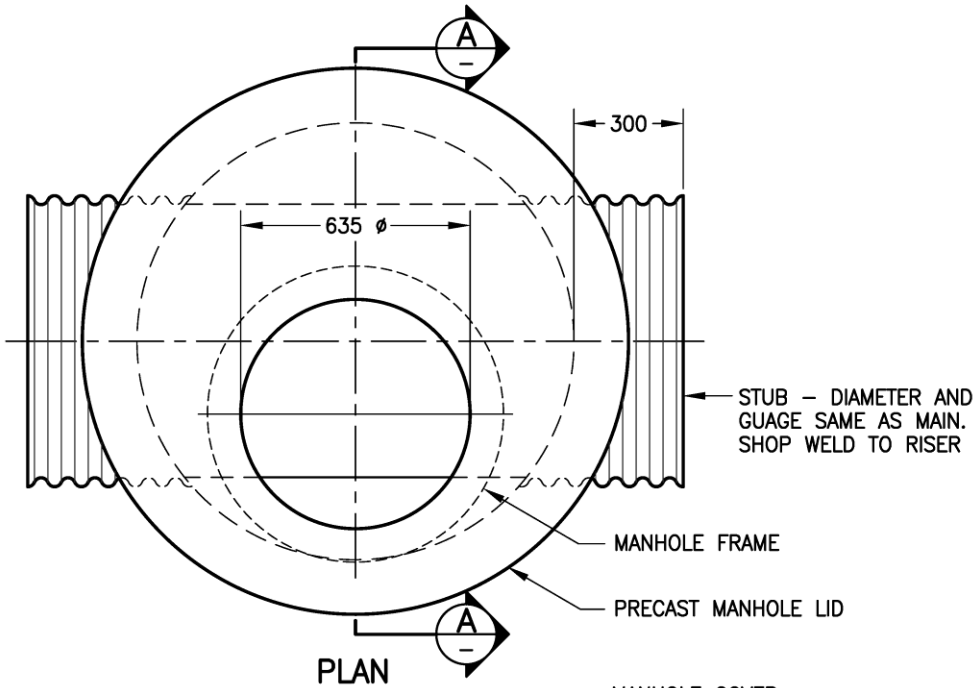
1. SUPPLIER TO INSTALL 19 ø GALVANIZED LADDER RUNGS OR PREAPPROVED ALTERNATIVE AS PER SS 145
2. CONCRETE PIPE TO CONFORM TO ASTM C 76.
3. ALL PRECAST WORK TO CONFORM TO ASTM C478 OR CSA A257.
4. MANHOLE LID TO WITHSTAND HIGHWAY LIVE LOADING IN ACCORDANCE WITH THE APPLICABLE CLAUSES OF SS 582.
5. MANHOLE TEE SECTION SHALL BE BEDDED AS PER SUBSECTION 582.37.
6. MANHOLE TEE SECTIONS MAY BE ORDERED AS INCREASERS ON THE UPSTREAM END TO ALLOW FOR PIPE SIZE CHANGES.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES

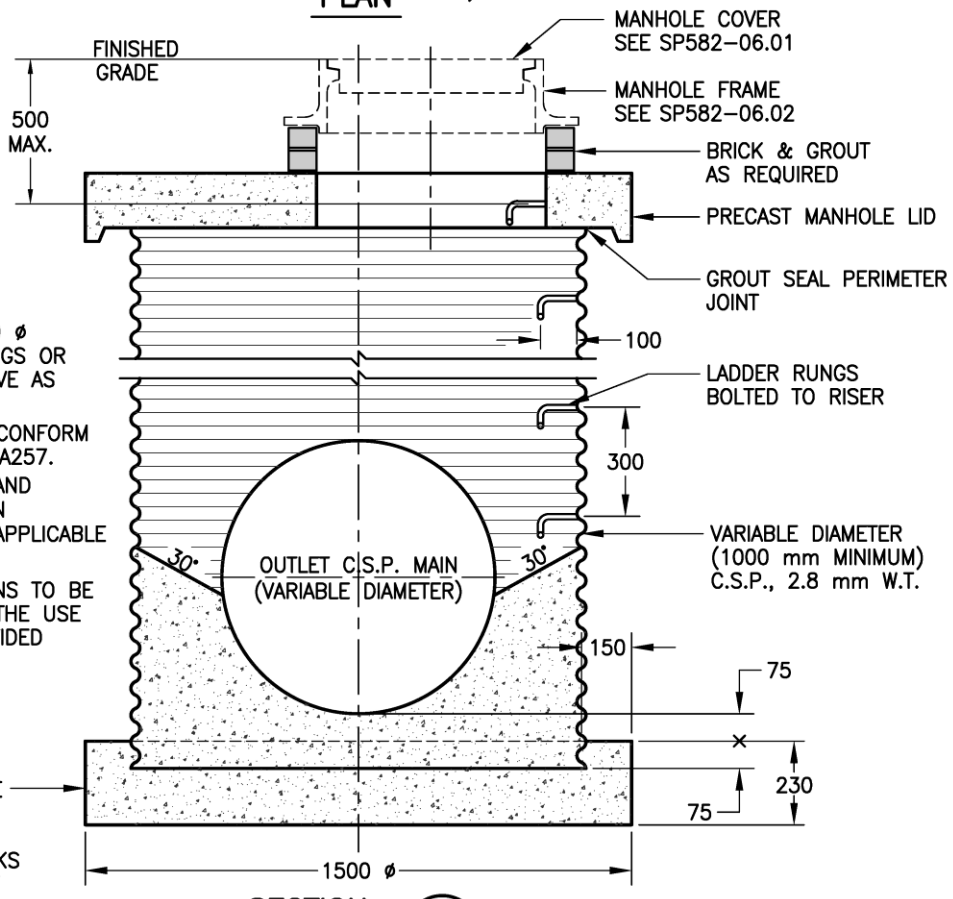
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

CORRUGATED STEEL MANHOLE MAIN SMALLER THAN RISER

SP582–03.07



PLAN



SECTION A

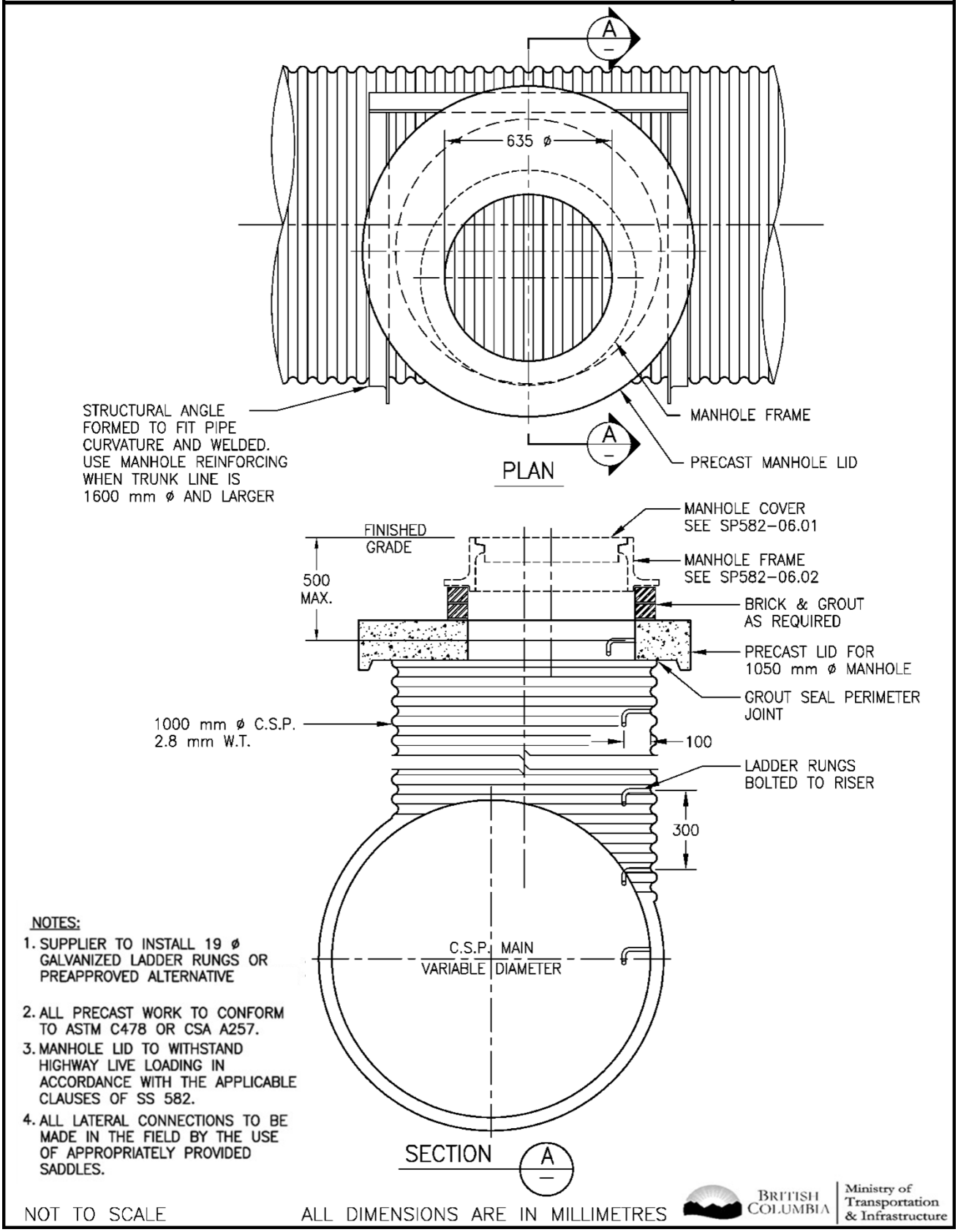
NOTES:

1. SUPPLIER TO INSTALL 19 ø GALVANIZED LADDER RUNGS OR PREAPPROVED ALTERNATIVE AS PER SS 145
2. ALL PRECAST WORK TO CONFORM TO ASTM C478 OR CSA A257.
3. MANHOLE LID TO WITHSTAND HIGHWAY LIVE LOADING IN ACCORDANCE WITH THE APPLICABLE CLAUSES OF SS 582.
4. ALL LATERAL CONNECTIONS TO BE MADE IN THE FIELD BY THE USE OF APPROPRIATELY PROVIDED SADDLES.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

CORRUGATED STEEL MANHOLE MAIN LARGER THAN RISER	SP582–03.08



NOTES:

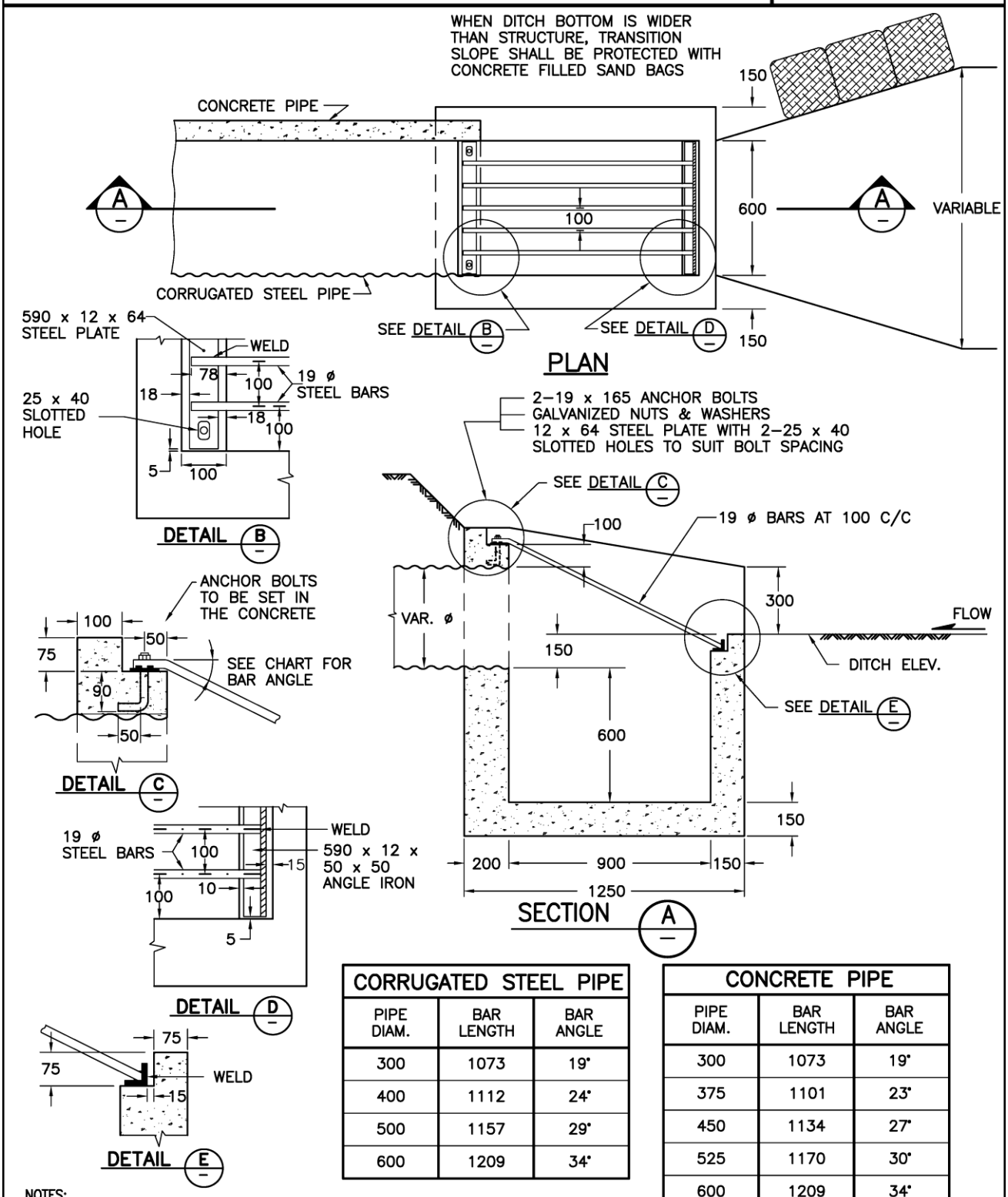
1. SUPPLIER TO INSTALL 19 ø GALVANIZED LADDER RUNGS OR PREAPPROVED ALTERNATIVE
2. ALL PRECAST WORK TO CONFORM TO ASTM C478 OR CSA A257.
3. MANHOLE LID TO WITHSTAND HIGHWAY LIVE LOADING IN ACCORDANCE WITH THE APPLICABLE CLAUSES OF SS 582.
4. ALL LATERAL CONNECTIONS TO BE MADE IN THE FIELD BY THE USE OF APPROPRIATELY PROVIDED SADDLES.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

CAST IN PLACE INLET STRUCTURE

SP582–04.01



CORRUGATED STEEL PIPE		
PIPE DIAM.	BAR LENGTH	BAR ANGLE
300	1073	19°
400	1112	24°
500	1157	29°
600	1209	34°

CONCRETE PIPE		
PIPE DIAM.	BAR LENGTH	BAR ANGLE
300	1073	19°
375	1101	23°
450	1134	27°
525	1170	30°
600	1209	34°

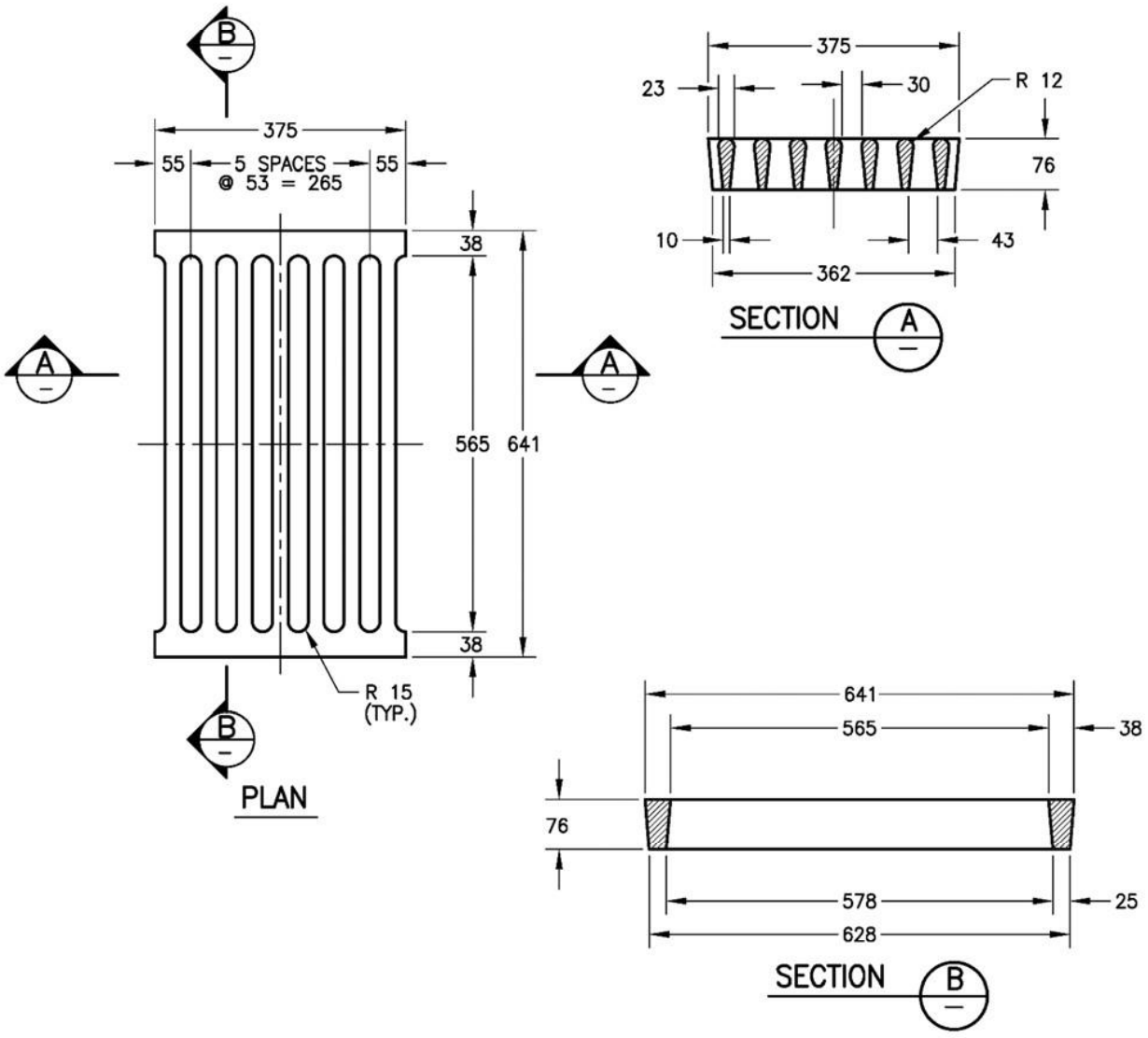
NOTES:
1. ALL STEELWORK TO CONFORM TO CSA SPECIFICATION G40.21M GRADE 300W.
2. ALL STEELWORK TO BE GALVANIZED AFTER FABRICATION IN ACCORDANCE WITH ASTM C123 OR C153, AS APPROPRIATE.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

TYPE 'A' CATCH BASIN GRATE (FREEWAY GRATE)

SP582–05.01



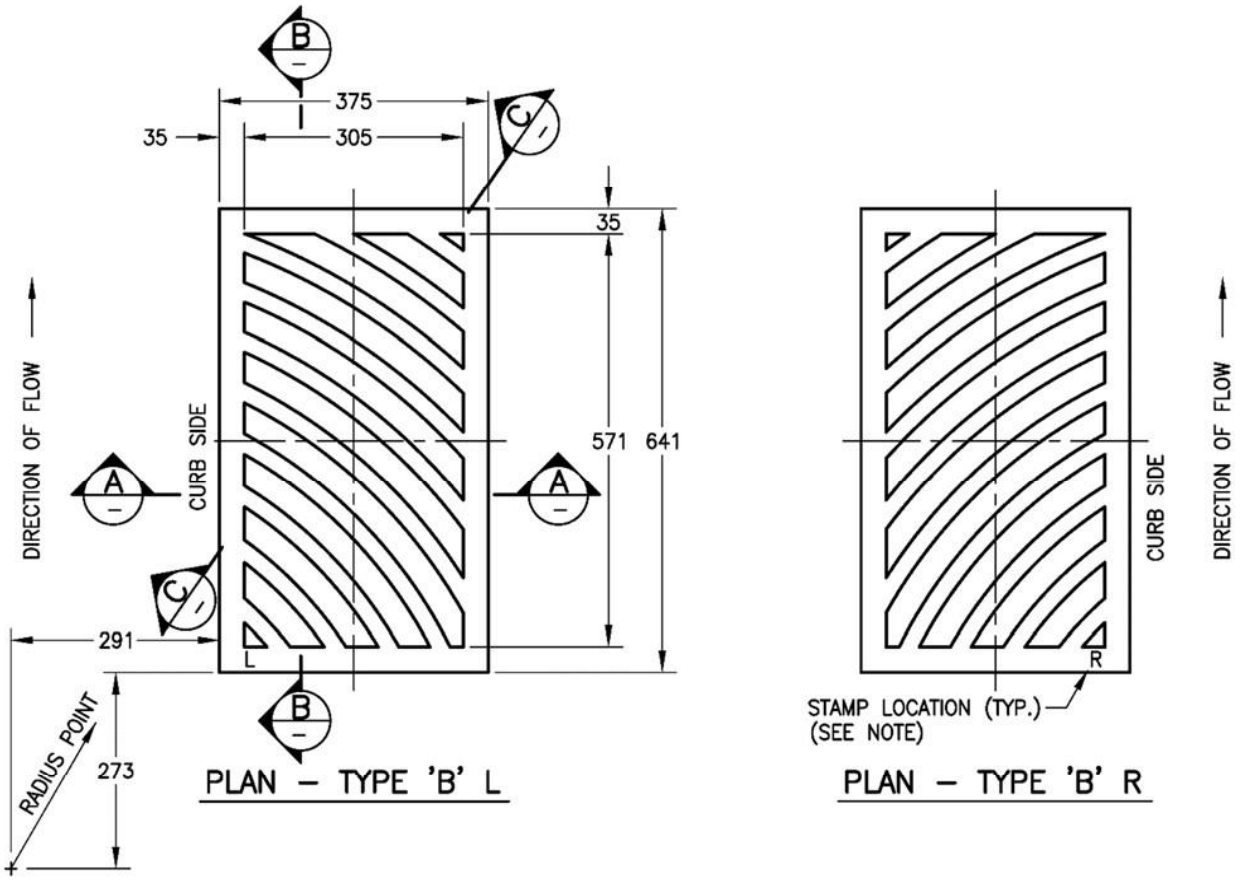
NOTES:
1. GRATES TO FIT CATCH BASIN FRAMES (SEE SP582–05.03 & SP582–05.04).
2. ALL MATERIAL TO BE CAST IRON.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

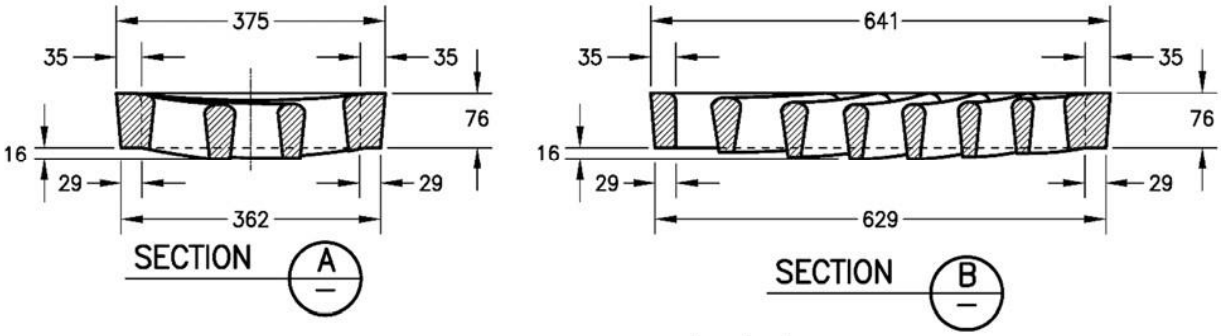
TYPE 'B' CATCH BASIN GRATE (BICYCLE SAFE GRATE)

SP582–05.02



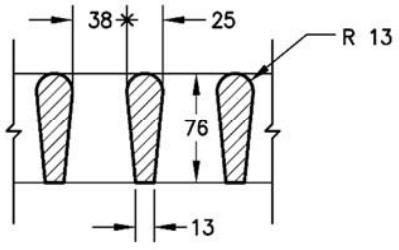
PLAN – TYPE 'B' L

PLAN – TYPE 'B' R



SECTION A

SECTION B



SECTION C

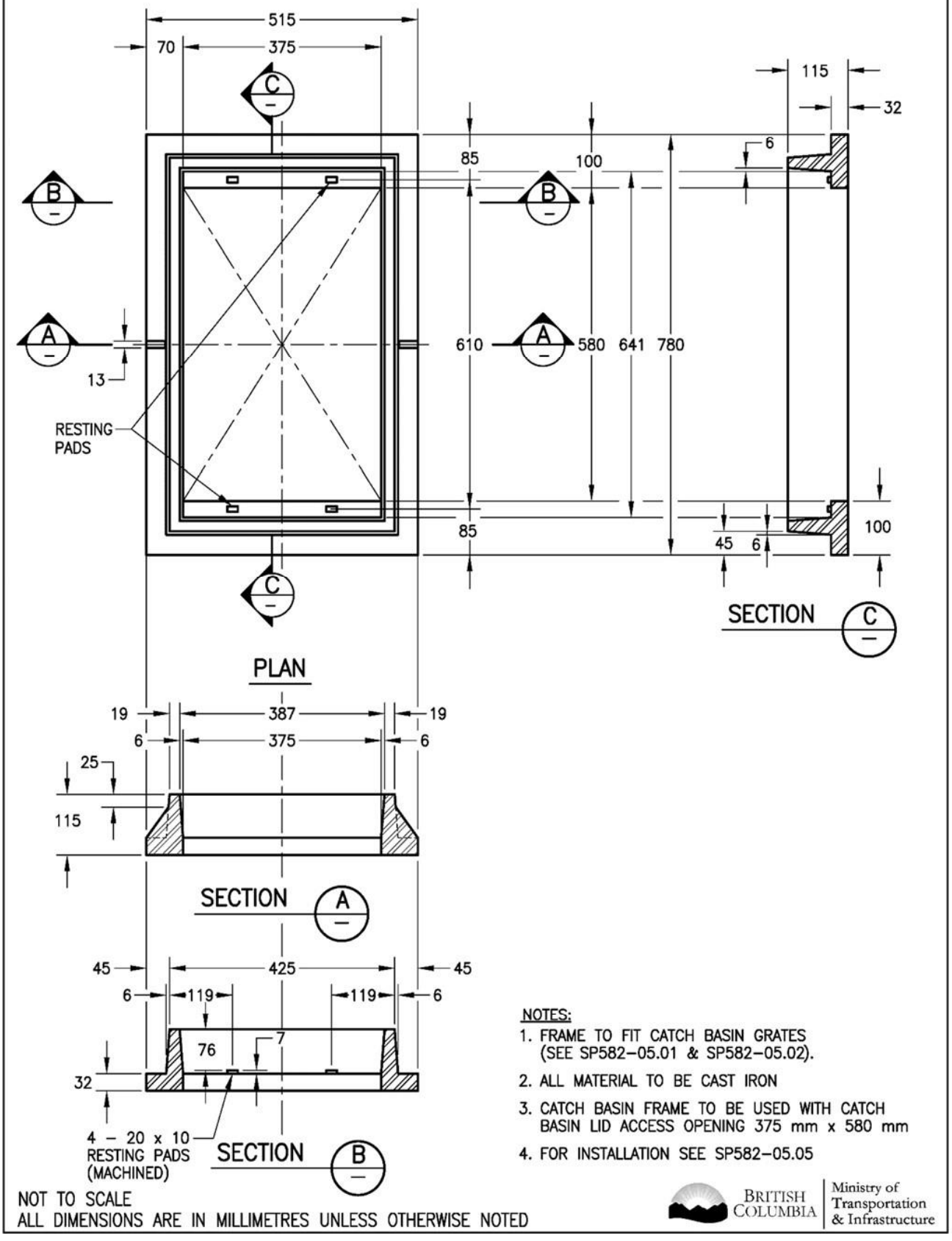
NOTES:
1. GRATES TO FIT CATCH BASIN FRAMES (SEE SP582–05.03 & SP582–05.04).
2. ALL MATERIAL TO BE CAST IRON.
3. GRATES TO BE STAMPED 'L' OR 'R' AS APPLICABLE, AT TIME OF MANUFACTURE.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

SINGLE INLET CATCH BASIN FRAME

SP582–05.03



NOTES:

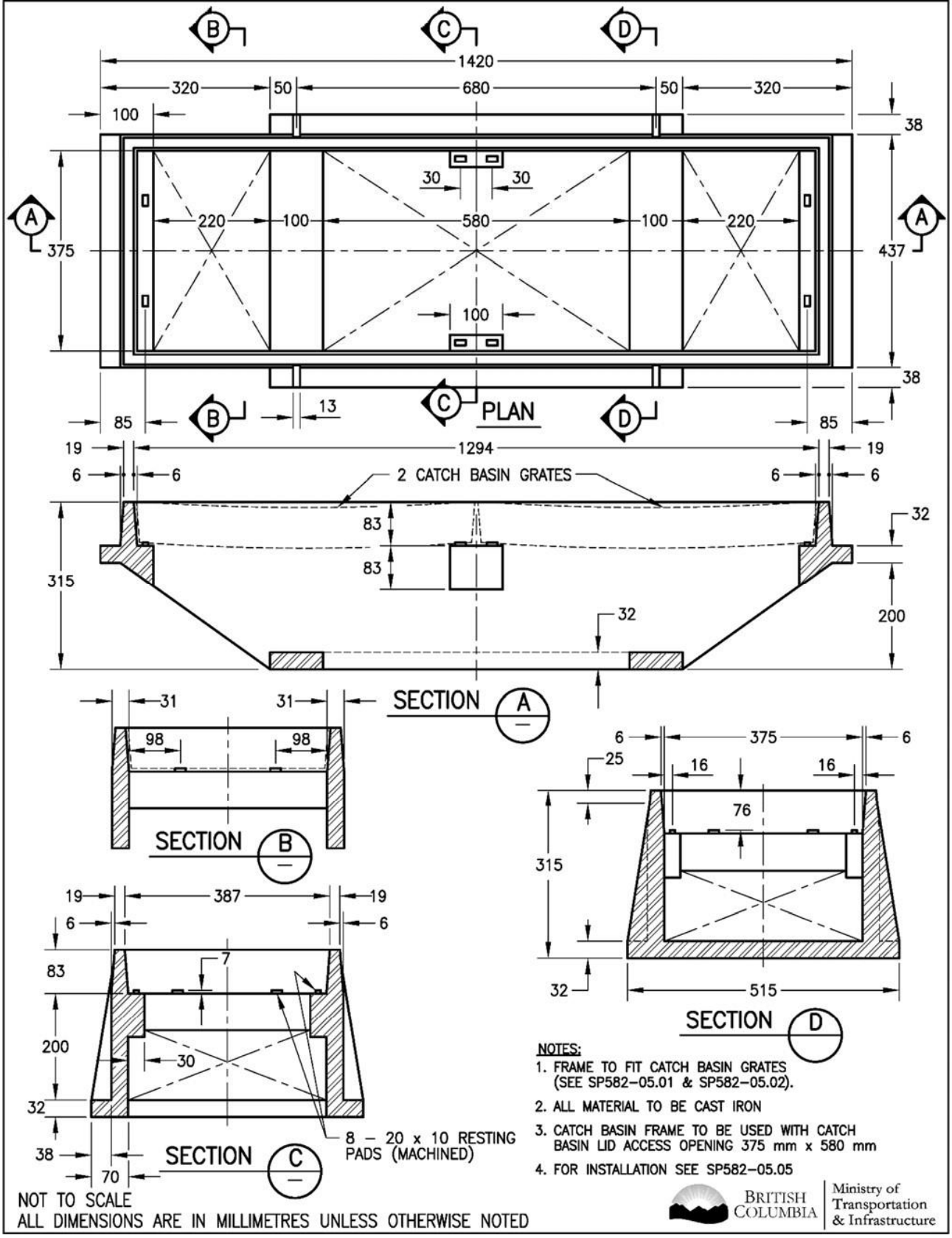
1. FRAME TO FIT CATCH BASIN GRATES (SEE SP582–05.01 & SP582–05.02).
2. ALL MATERIAL TO BE CAST IRON
3. CATCH BASIN FRAME TO BE USED WITH CATCH BASIN LID ACCESS OPENING 375 mm x 580 mm
4. FOR INSTALLATION SEE SP582–05.05

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

TWIN INLET CATCH BASIN FRAME

SP582–05.04

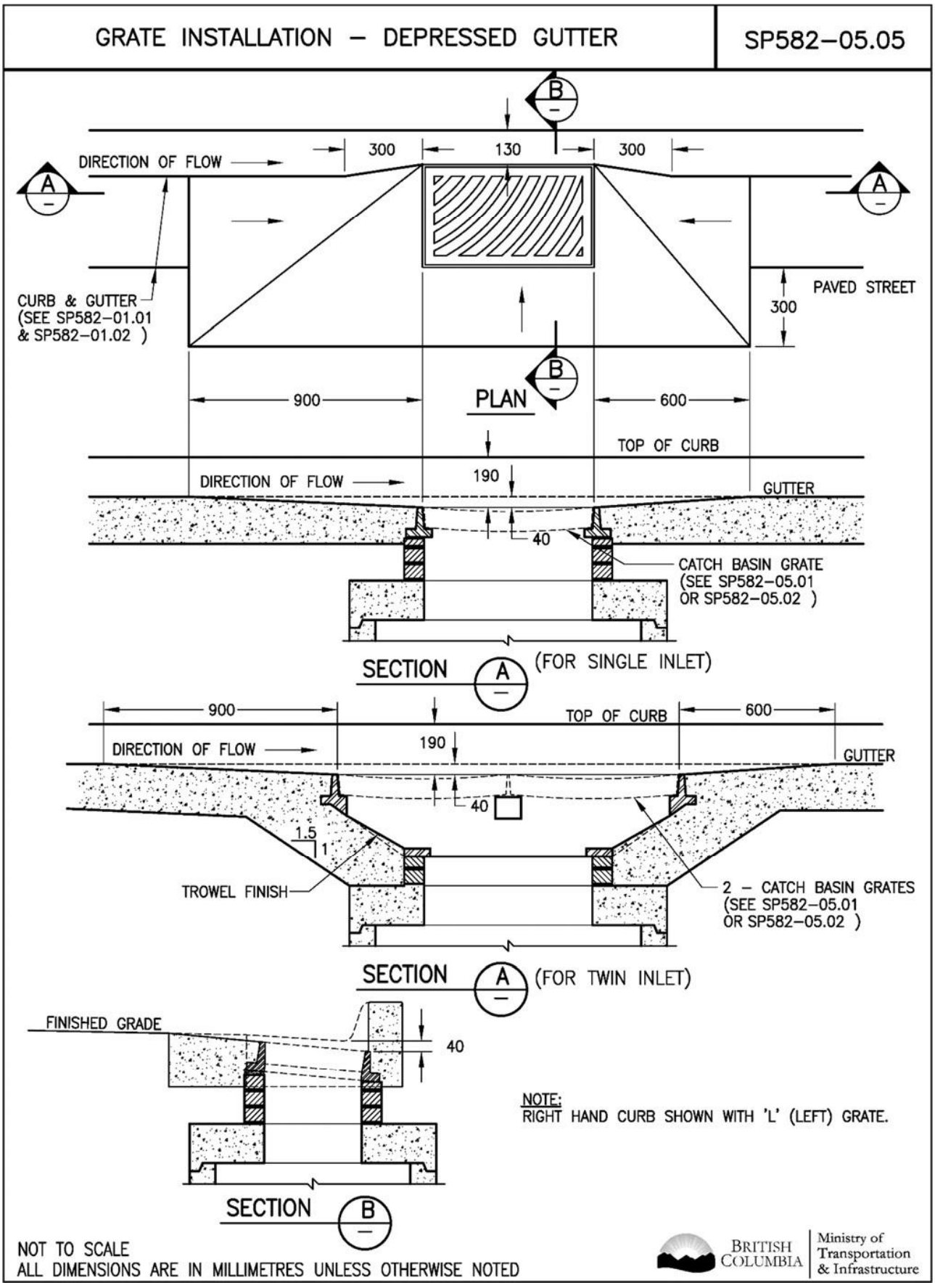


NOTES:

1. FRAME TO FIT CATCH BASIN GRATES (SEE SP582–05.01 & SP582–05.02).
2. ALL MATERIAL TO BE CAST IRON
3. CATCH BASIN FRAME TO BE USED WITH CATCH BASIN LID ACCESS OPENING 375 mm x 580 mm
4. FOR INSTALLATION SEE SP582–05.05

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

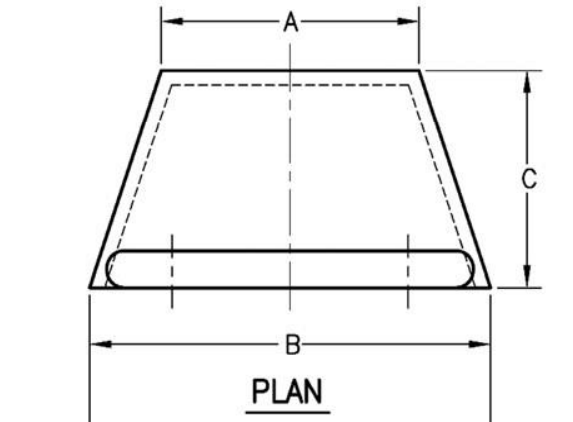
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure


GRATE INSTALLATION – DEPRESSED GUTTER
SP582–05.05
DIRECTION OF FLOW
300
130
300
CURB & GUTTER
(SEE SP582–01.01
& SP582–01.02)
PAVED STREET
300
900
PLAN
600
TOP OF CURB
DIRECTION OF FLOW
190
GUTTER
40
CATCH BASIN GRATE
(SEE SP582–05.01
OR SP582–05.02)
SECTION A (FOR SINGLE INLET)
900
TOP OF CURB
600
DIRECTION OF FLOW
190
GUTTER
40
1.5
1
TROWEL FINISH
2 – CATCH BASIN GRATES
(SEE SP582–05.01
OR SP582–05.02)
SECTION A (FOR TWIN INLET)
FINISHED GRADE
40
NOTE:
RIGHT HAND CURB SHOWN WITH 'L' (LEFT) GRATE.
SECTION B
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED
BRITISH COLUMBIA
Ministry of Transportation & Infrastructure

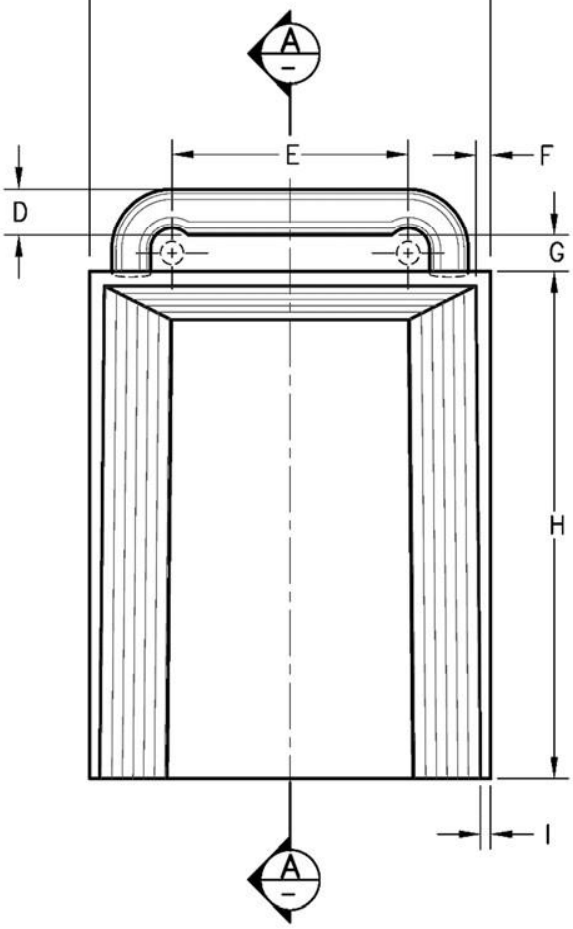
TRAPPING HOOD

SP582–05.06

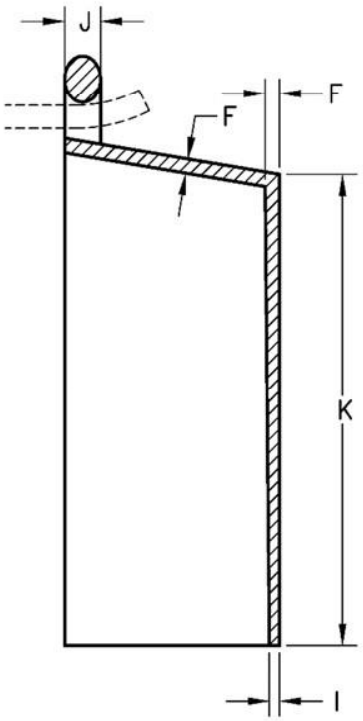
DIM.	1A 150 OUTLET	9 200 OUTLET	9A 300 OUTLET
A	180	215	320
B	280	345	445
C	150	215	255
D	32	32	38
E	165	190	305
F	10	13	13
G	25	25	25
H	350	425	535
I	7	10	10
J	25	29	29
K	325	370	470
MASS	16 kg	25 kg	49 kg



PLAN



ELEVATION



SECTION A

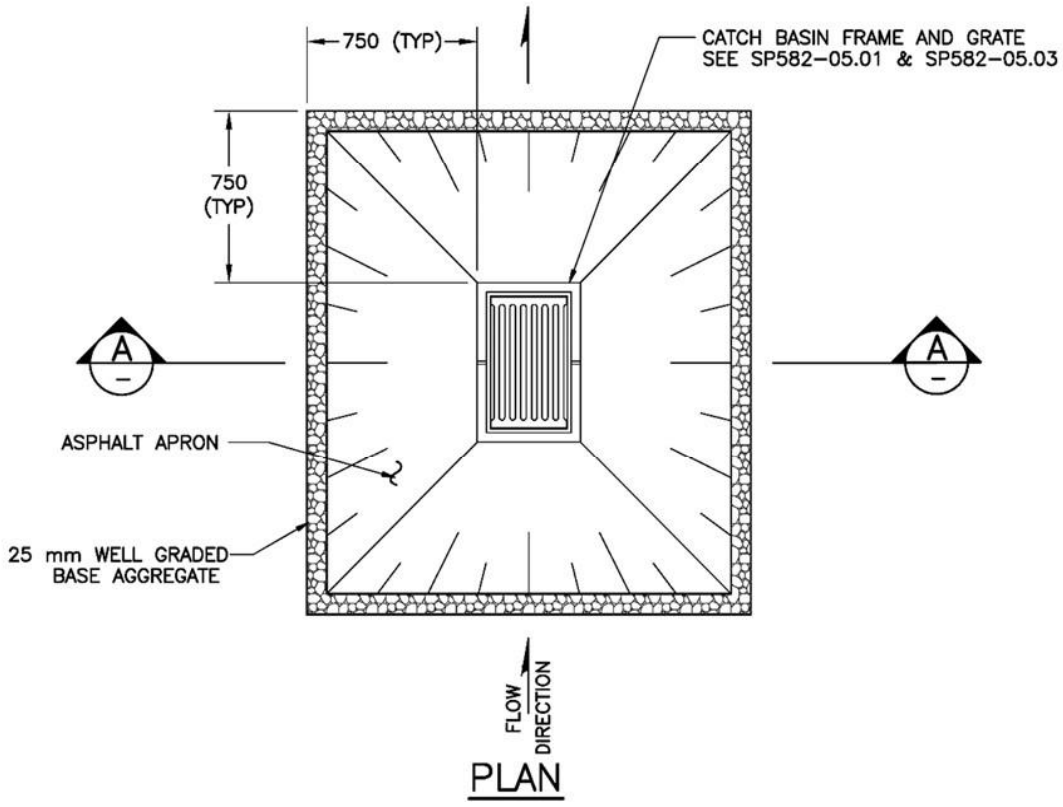
NOTE:
ALL MATERIAL TO BE CAST IRON.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

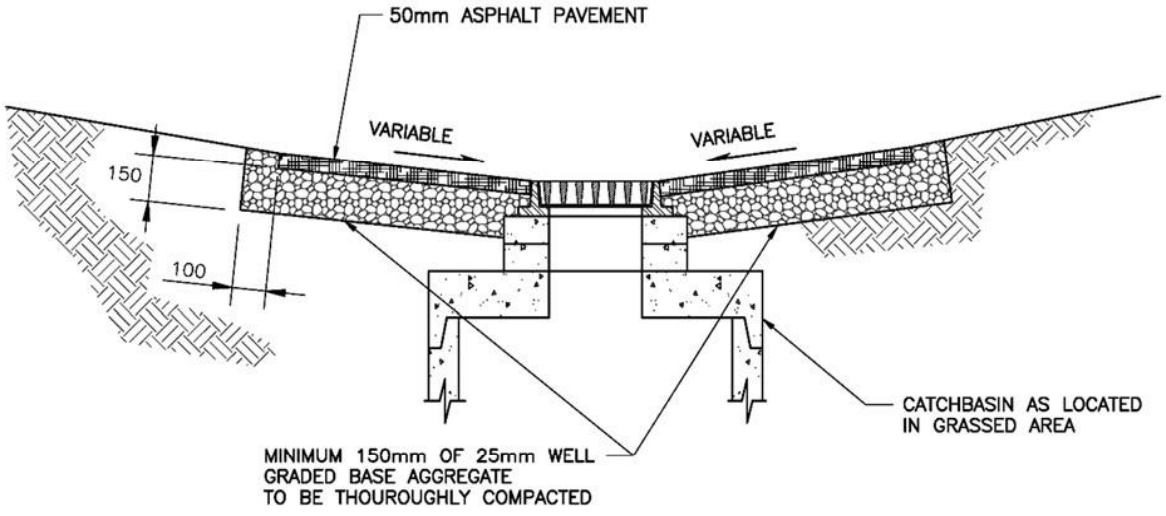
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

ASPHALT APRON

SP582–05.07



PLAN



SECTION A

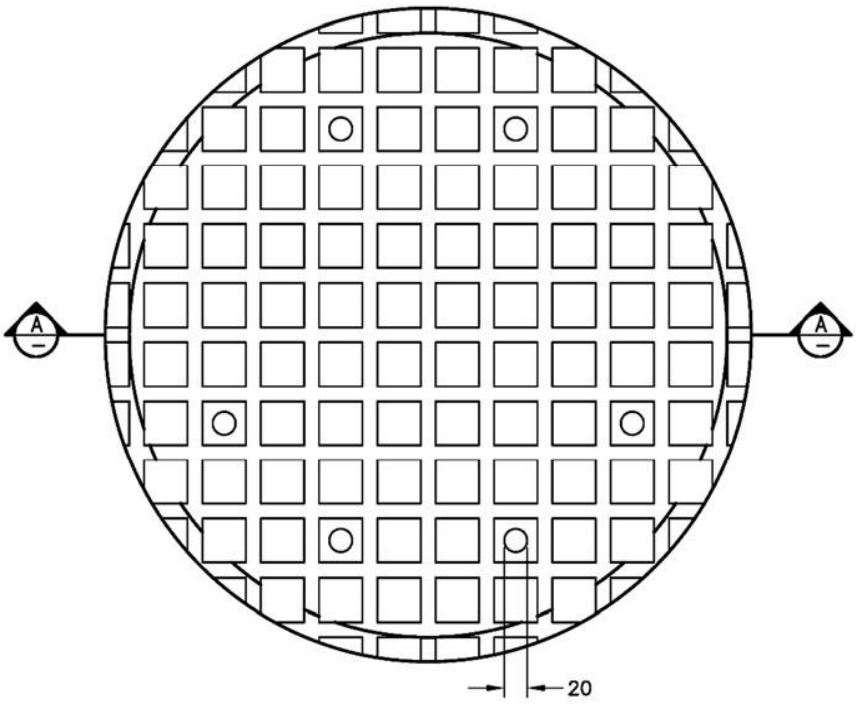
NOTE:
FOR DETAILS OF CATCH BASIN
SEE SP582–02.01 THROUGH SP582–02.05.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

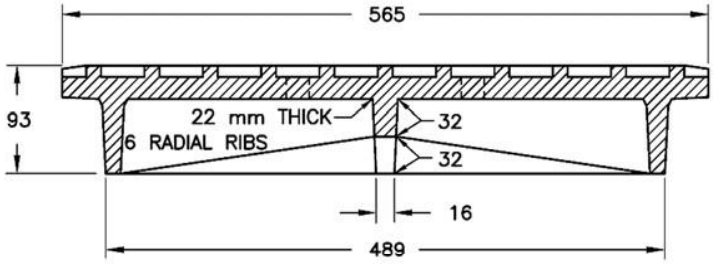
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

MANHOLE COVER

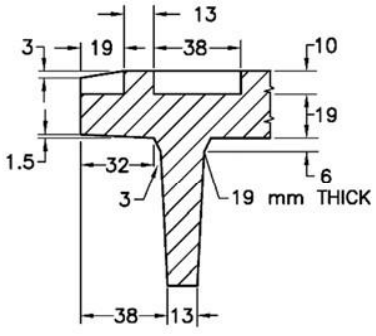
SP582–06.01



MATERIAL: CAST IRON



SECTION A



DETAIL OF RIM

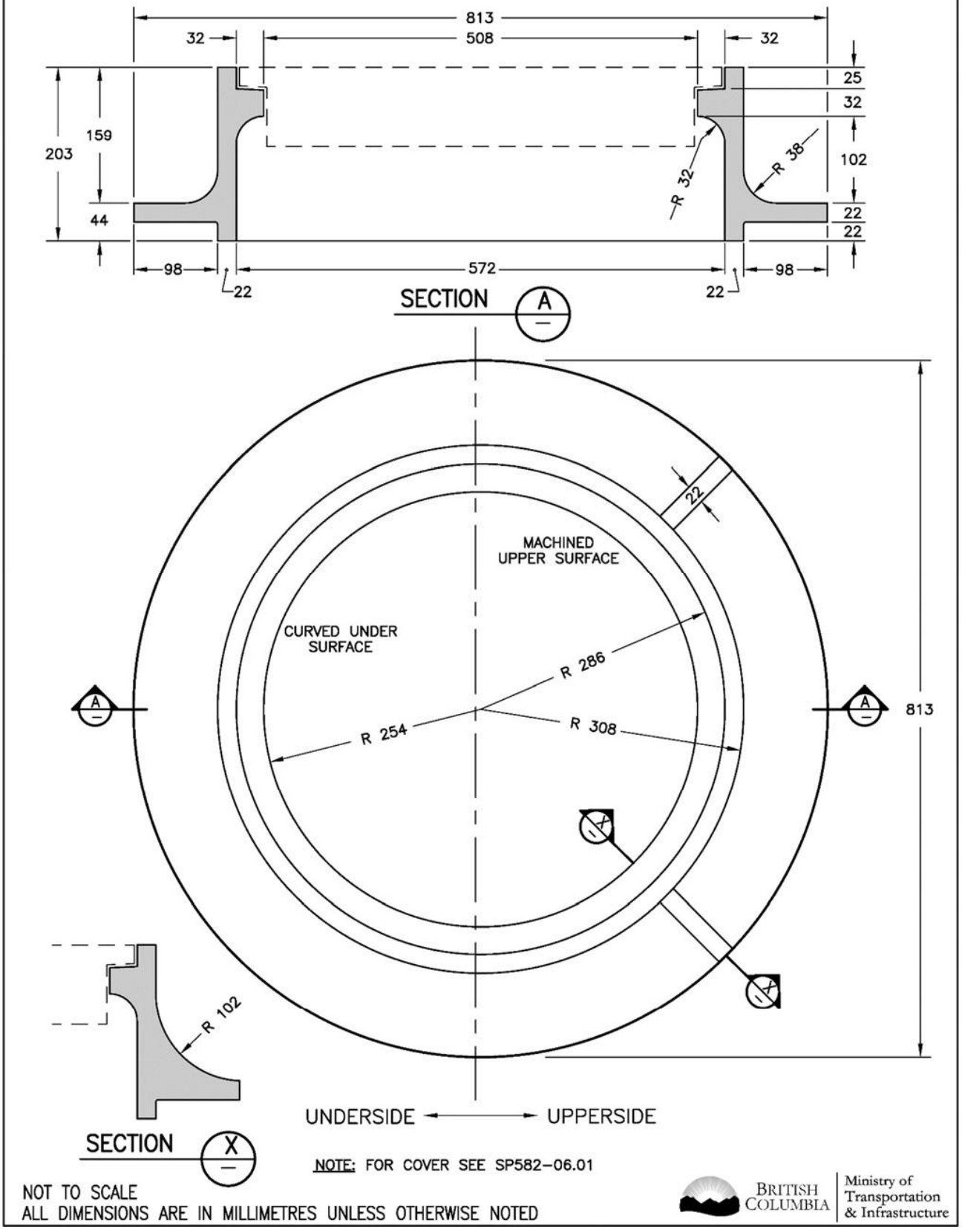
NOTE: FOR FRAME SEE SP582–06.02

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

MANHOLE FRAME

SP582–06.02



NOTE: FOR COVER SEE SP582–06.01

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

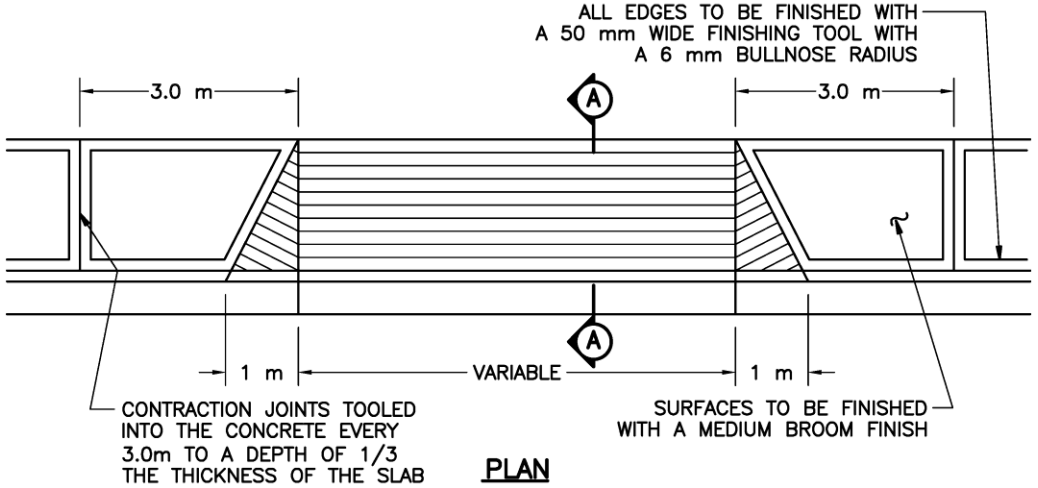
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

CONCRETE SIDEWALK & DRIVEWAY ENTRANCE

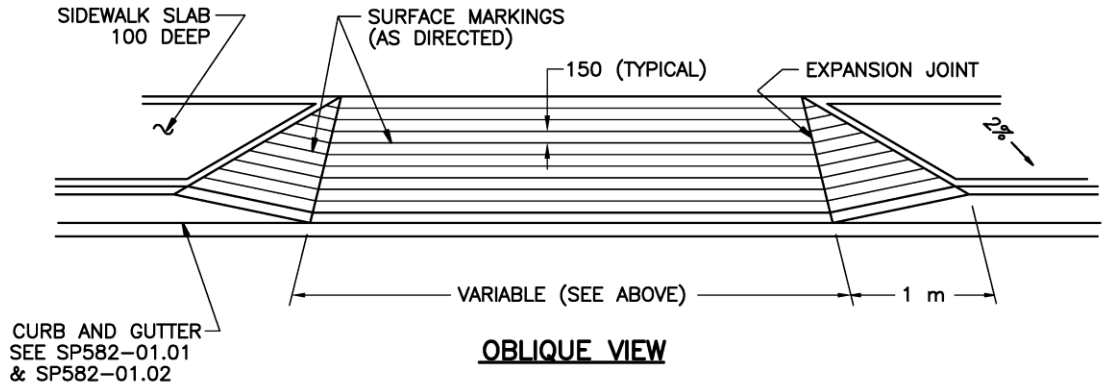
SP582–07.01

EXPANSION JOINTS
- AT 15 m INTERVALS ACROSS SIDEWALK.
- AROUND UTILITY POLES, CONCRETE STRUCTURES AND BUILDINGS.
- AT EACH END OF THICKENED SIDEWALK SECTION AND AT MID–SPAN WHEN THE LENGTH IS GREATER THAN 9 m.

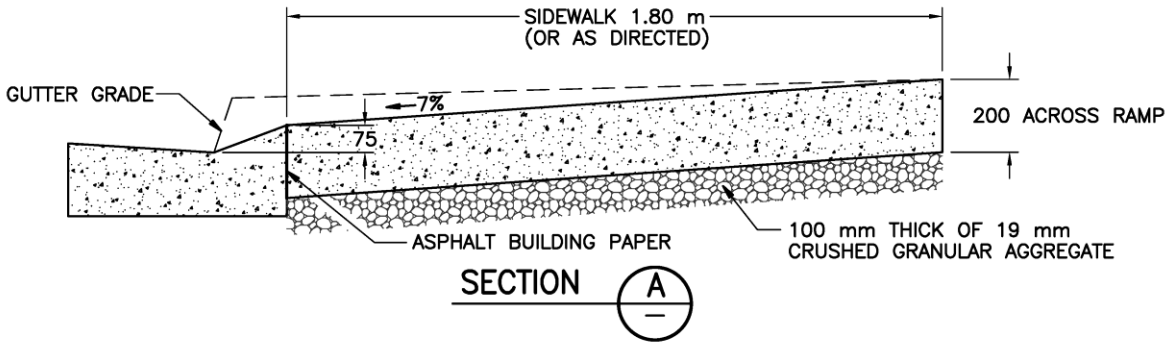
DRIVEWAY WIDTHS
- COMMERCIAL 2 WAY TRAFFIC 7.5 – 15 m
- COMMERCIAL 1 WAY TRAFFIC 6 – 10 m
- RESIDENTIAL.....6 m



PLAN



OBLIQUE VIEW



SECTION A–A

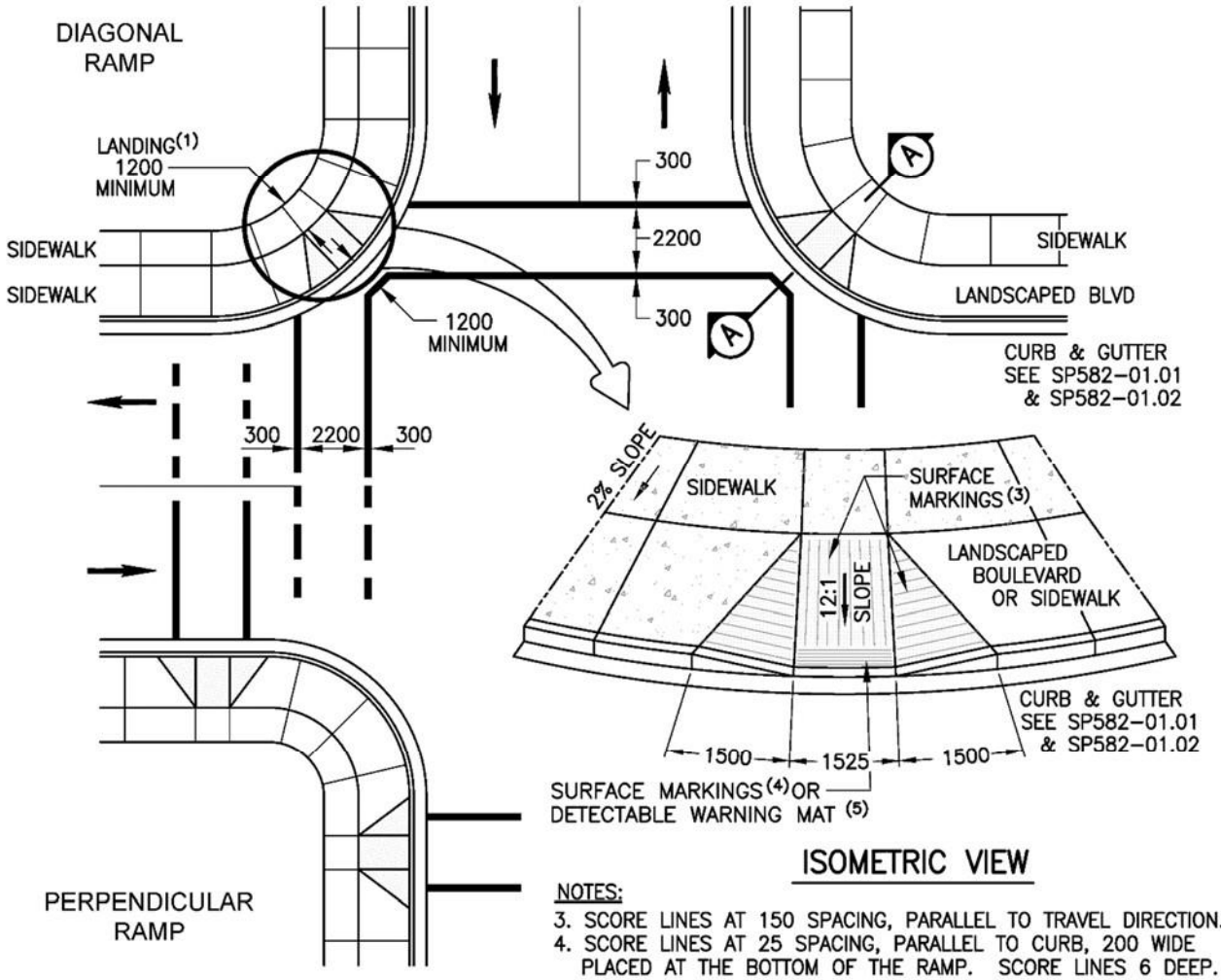
NOTES:
1. FOR OPEN SHOULDER DRIVEWAY DETAIL SEE B.C. SUPPLEMENT TO TAC.
2. ADD 150 mm TO WIDTH WHEN SIDEWALK FENCE IS REQUIRED.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

SIDEWALK RAMP – DIAGONAL AND PERPENDICULAR

SP582–07.02.01

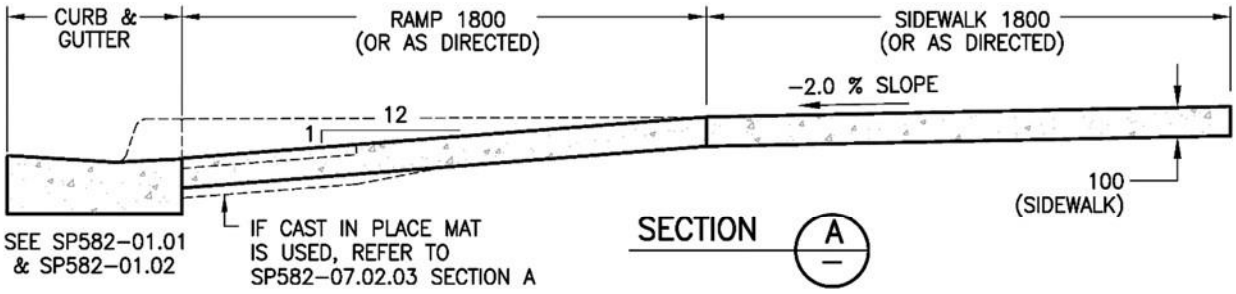


PLAN

ISOMETRIC VIEW

NOTES:

3. SCORE LINES AT 150 SPACING, PARALLEL TO TRAVEL DIRECTION.
4. SCORE LINES AT 25 SPACING, PARALLEL TO CURB, 200 WIDE PLACED AT THE BOTTOM OF THE RAMP. SCORE LINES 6 DEEP.
5. DETECTABLE WARNING MATS SHALL BE 610 (24") DEEP AND THE FULL WIDTH OF THE CURB RAMP. THESE MATS ARE TYPICALLY MANUFACTURED WITH IMPERIAL DIMENSIONS IN 12" INCREMENTS; THEREFORE, CONSTRUCT THE CURB RAMP WIDTH TO SUIT.



SECTION A

NOTE:

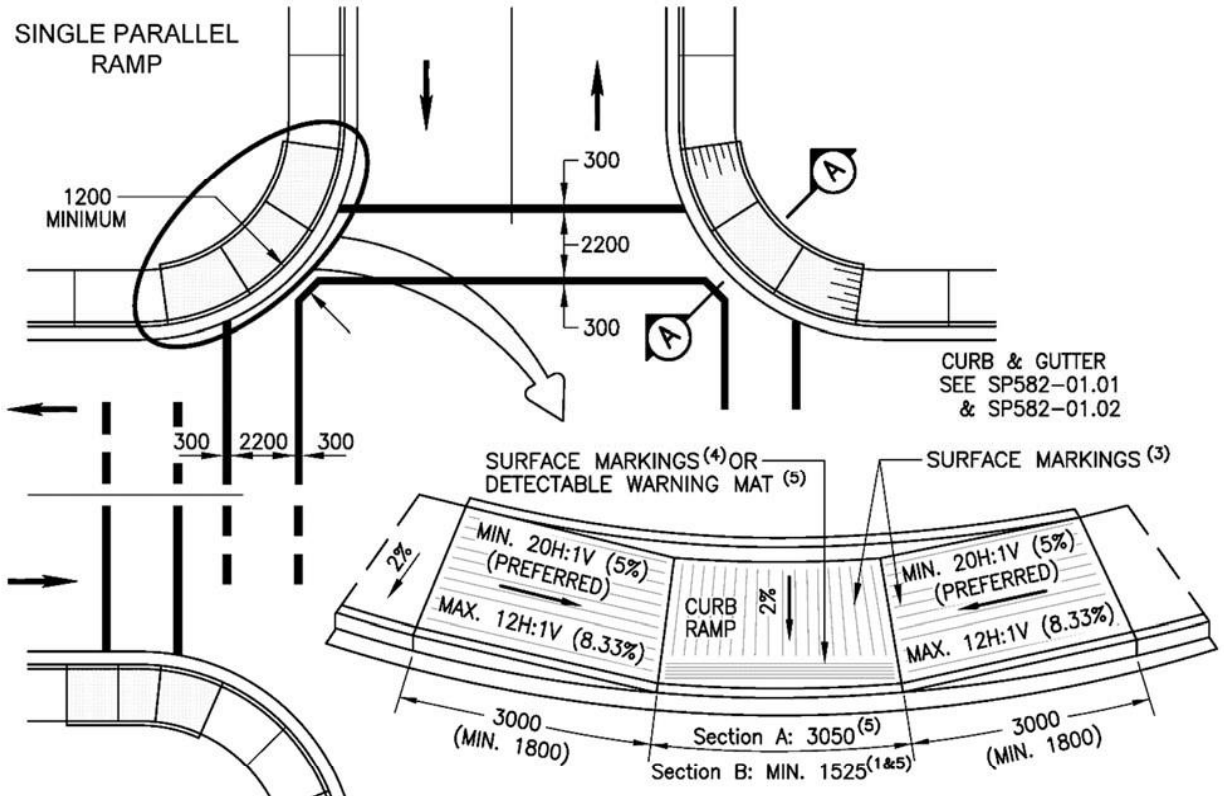
1. PERPENDICULAR RAMPS ARE PREFERRED. IF IT IS NOT POSSIBLE TO PROVIDE A TOP LEVEL LANDING, DIAGONAL AND PERPENDICULAR RAMPS SHOULD NOT BE INSTALLED; USE SP582–07.02.02 PARALLEL RAMPS.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

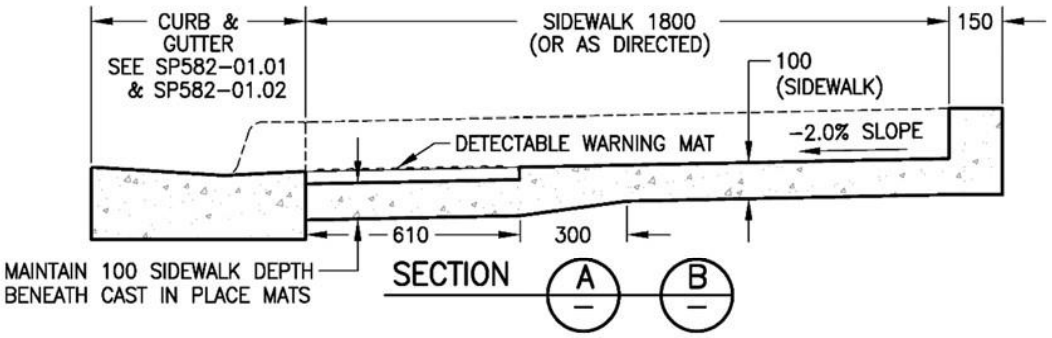
SIDEWALK RAMP – PARALLEL

SP582–07.02.02



NOTES:

3. SCORE LINES AT 150 SPACING, PARALLEL TO TRAVEL DIRECTION.
4. SCORE LINES AT 25 SPACING, PARALLEL TO CURB, 200 WIDE PLACED AT THE BOTTOM OF THE RAMP. SCORE LINES 6 DEEP.
5. DETECTABLE WARNING MATS SHALL BE 610 (24") DEEP AND THE FULL WIDTH OF THE CURB RAMP. THESE MATS ARE TYPICALLY MANUFACTURED WITH IMPERIAL DIMENSIONS IN 12" INCREMENTS; THEREFORE, CONSTRUCT THE CURB RAMP WIDTH TO SUIT.



NOTE:

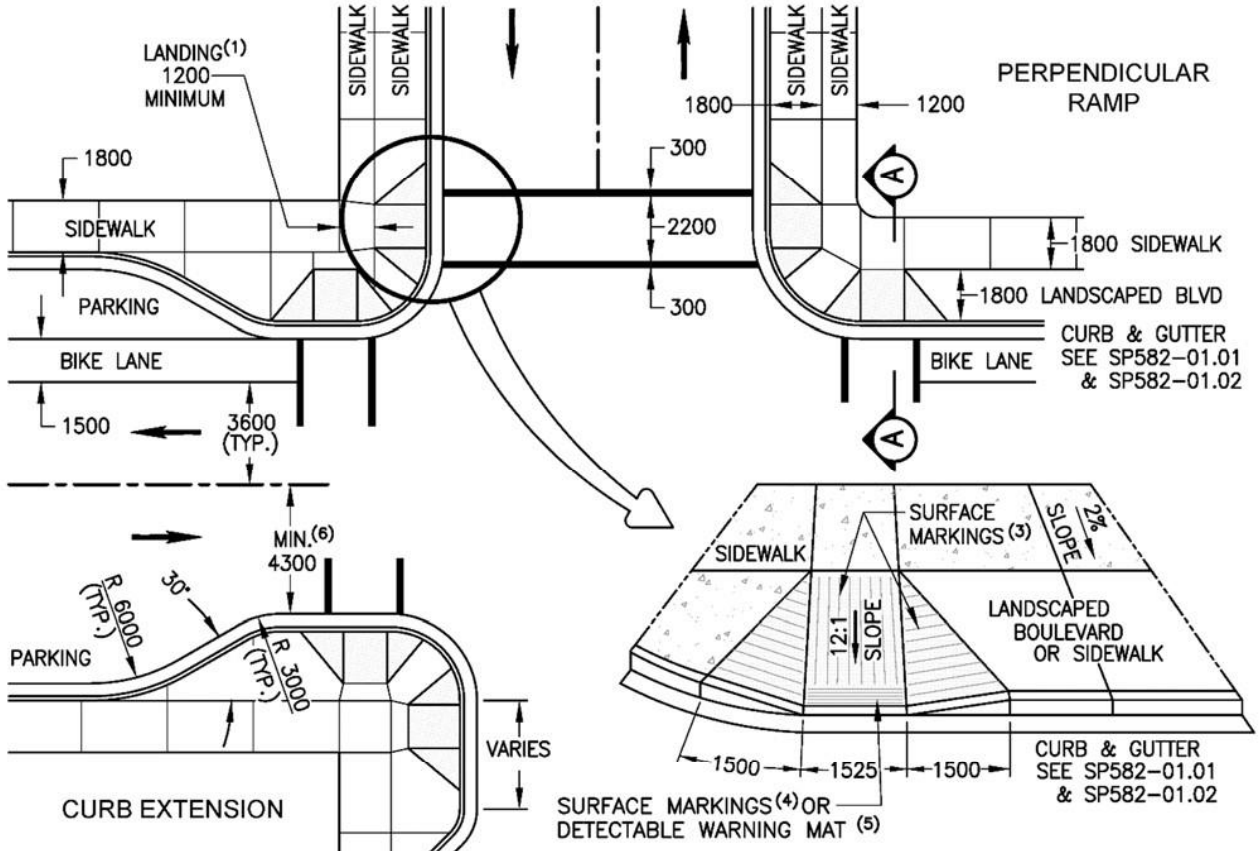
1. FOR SECTION B RAMPS, MINIMUM WIDTH OF RAMP IS 1525. IT MAY BE NECESSARY TO BUILD WIDER RAMPS IN BUSY URBAN AREAS WHERE THE VOLUME OF PEDESTRIAN TRAFFIC IS HIGH.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

SIDEWALK RAMP – CURB EXTENSION

SP582–07.02.03



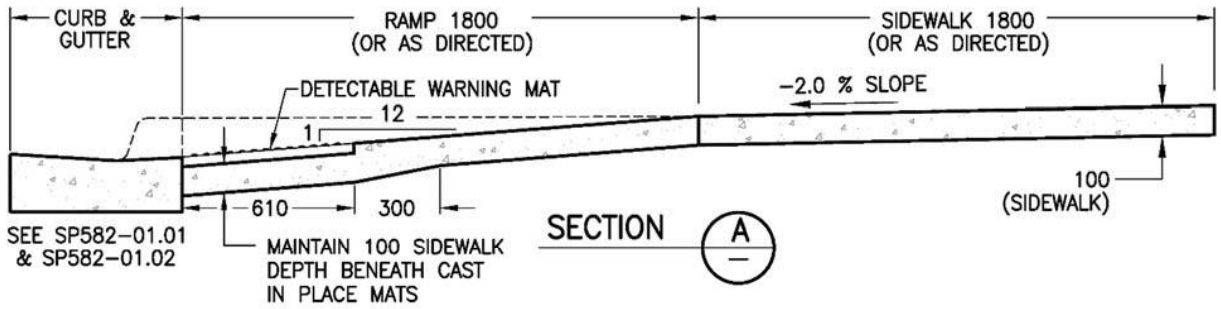
NOTES:

6. SHARED ROADWAY WIDTH WHERE BIKE LANES ARE NOT PROVIDED.

PLAN

NOTES:

3. SCORE LINES AT 150 SPACING, PARALLEL TO TRAVEL DIRECTION.
4. SCORE LINES AT 25 SPACING, PARALLEL TO CURB, 200 WIDE PLACED AT THE BOTTOM OF THE RAMP. SCORE LINES 6 DEEP.
5. DETECTABLE WARNING MATS SHALL BE 610 (24") DEEP AND THE FULL WIDTH OF THE CURB RAMP. THESE MATS ARE TYPICALLY MANUFACTURED WITH IMPERIAL DIMENSIONS IN 12" INCREMENTS; THEREFORE, CONSTRUCT THE CURB RAMP WIDTH TO SUIT.



NOTE:

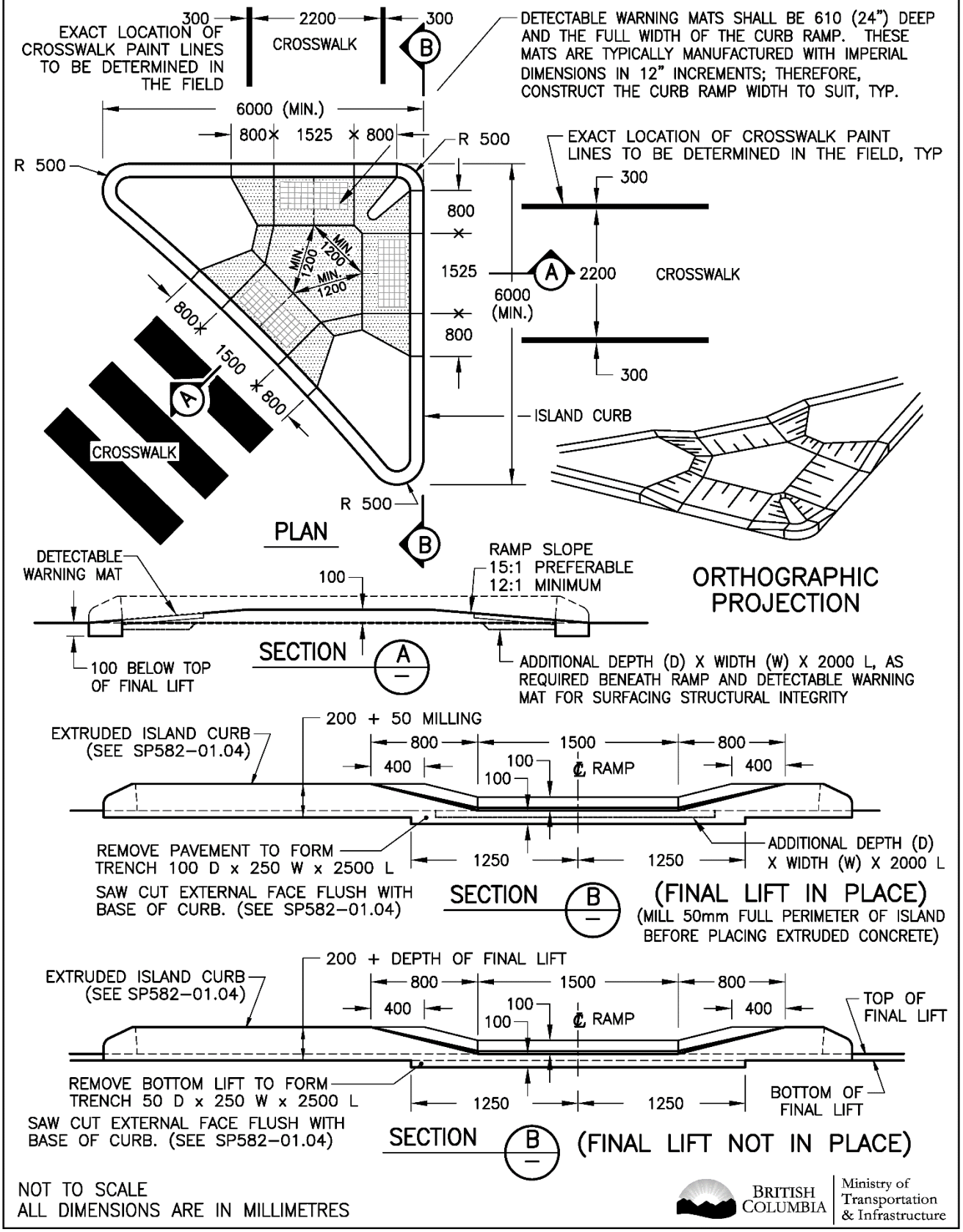
1. IF IT IS NOT POSSIBLE TO PROVIDE A TOP LEVEL LANDING, PERPENDICULAR RAMPS SHOULD NOT BE INSTALLED; USE SP582–07.02.02 PARALLEL RAMPS.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

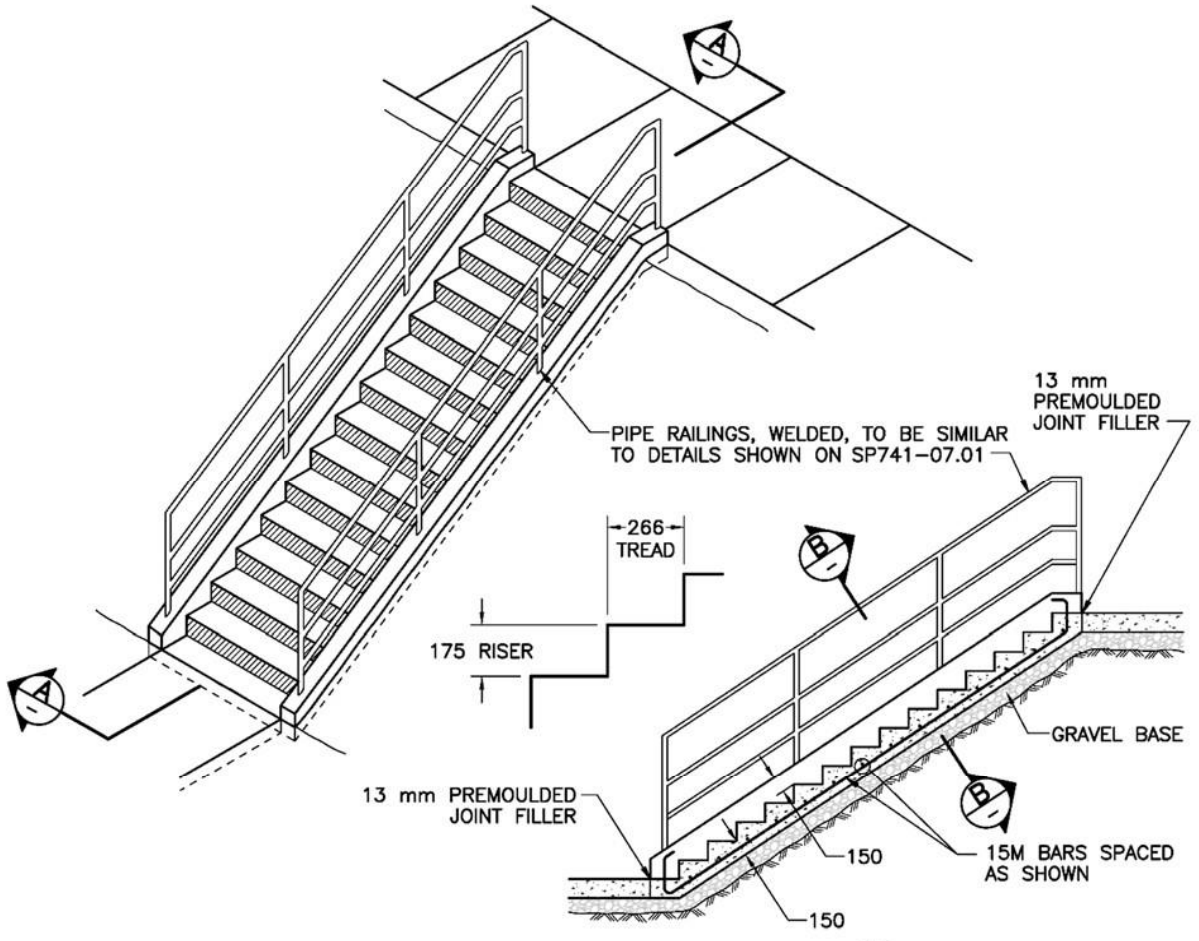
ISLAND RAMP

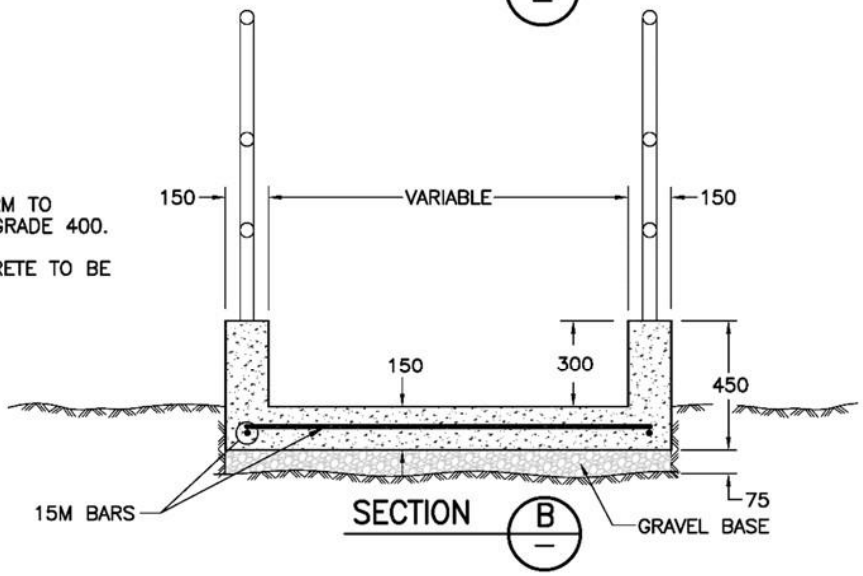
SP582–07.03



REINFORCED CONCRETE STEPS

SP582–08.01





NOTES:

1. REINFORCING STEEL TO CONFORM TO CSA SPECIFICATION G30.18–M GRADE 400.
2. ALL EXPOSED EDGES OF CONCRETE TO BE ROUNDED TO 13 mm RADIUS.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

SECTION 586

PLANT MIXED
PORTLAND CEMENT TREATED BASE COURSE

DESCRIPTION

586.01 Preliminary and General – This Section describes the materials, plant equipment and quality of work required for the stabilization of base course material with Portland cement.

586.02 General Description of Work – The cement treated base shall consist of prepared aggregate, Portland cement and water uniformly mixed to meet the criteria specified, in a central mixing plant of satisfactory type, hauled out and laid on the prepared subgrade and compacted to established line and grade. The Contractor shall supply all materials.

MATERIALS

586.11 Cement – Portland Cement to be supplied for the work will comply with the requirements of SS 211 and tests of the CSA A3000 "Cementitious Materials Compendium".

586.12 Water – The water shall be clean and free from substances deleterious to the hardening of the cement stabilized base course.

586.13 Aggregate – The aggregate supplied shall be composed of inert durable material uniform in quality and free from organic impurities and soft or disintegrated particles. When tested according to ASTM C117, the aggregate shall meet the gradation requirements for 25 mm WGB as listed in Section 202.

The material shall also be tested in accordance with ASTM D4318 and shall have:

(a) a liquid limit not greater than twenty-five (≤ 25); and

(b) a plasticity index not greater than six (≤ 6).

All aggregate shall be stockpiled in horizontal layers spread and levelled to not more than 1.0 m in thickness. The stockpile area shall be cleared of all vegetation, trees, brush, rock and other debris and a uniform ground surface prepared before the aggregate is deposited on the site.

The material shall be loaded from the stockpile into the plant in such a manner as to ensure that a uniform cut of the pile is obtained.

Mixing may not commence until a minimum of 30% of the total quantities of aggregate required is in the stockpile.

Subsequent addition of prepared aggregate to the pile shall be organized to ensure continued uniformity of the aggregate supply to the mixing plant.

EQUIPMENT

586.21 Mixing Plant – The Portland Cement treated base course shall be mixed in a stationary plant employing either continuous or batch type equipment.

The plant shall be equipped with interlocked feeding and metering devices which will add the cement, water and aggregates in proportions established by the Ministry Representative to a tolerance of ± 3% by weight. The plant shall be capable of mixing the ingredients uniformly. Suitable storage shall be provided for bulk cement, and if required, bulk mineral filler. The capacity of the mixing plant shall be such that a minimum of 0.4 km of full width cement treated base can be laid per daily shift, i.e., sustained mixing capacity shall normally be not less than 250 tonnes per hour.

Details of the plant and its operation shall be submitted to the Ministry Representative for approval a minimum of seven (7) days prior to commencing construction.

586.22 Cement Treated Base Spreader – The spreading device may be self-propelled, pushed, or towed; and shall be so constructed and operated as to produce without segregation or surface tearing, a layer of material which is uniform in thickness and surface contour, either longitudinal or transverse, and free from irregularities in density. Whichever type of spreader may be used, it is mandatory that it be provided with a positive mechanical means for distributing the cement treated base transversely across the front of the cut-off blade or screed.

Spreading equipment which rides on the freshly spread material and produces tracks or partially compacted areas behind the cut-off blade or screed will be acceptable provided no displacement of material or filling of the tracks occurs, and provided further that the tracks are not of such depth as to be visible after compaction is completed. The use of a motor grader or any other spreading device, which requires repeated trips over the same area, will not be permitted for spreading.

586.23 Water Distributors – Water distributors shall be the pressure type, equipped with spray bar mounting nozzles similar to those used on asphalt distributors and capable of applying the water in accurate quantities and as a uniform spray. Splash plate type distributors or those equipped with spray bars that eject fine streams of water will not be permitted. The distributor must be provided with a satisfactorily accurate means of measuring the quantity of water sprayed.

CONSTRUCTION

586.31 Mixing – The mix proportions shall be determined by the Contractor, from samples taken at the commencement of aggregate production and in accordance with the mix design criteria provided in the Contract, and shall be subject to approval by the Ministry Representative.

The aggregate, mineral filler and cement shall be fed into the pugmill accurately, according to the proportions required, and shall be thoroughly dry mixed. Water shall then be added in the required amount and mixing continued until uniformity is achieved.

586.32 586.32 Spreading and Compaction

586.32.01 Surface – The surface of the subgrade shall be thoroughly moistened just prior to placing the cement treated base mixture thereon and shall be kept moist but not excessively wet until covered by the mixture.

586.32.02 Cement Treated Base Mixture – The cement treated base mixture shall be hauled to the roadbed in trucks equipped with protective covers and shall then be laid using a spreading device as per SS 586.22 to produce a layer of uniform density and cross section and in sufficient quantity to provide a compacted base conforming to the grade and cross section set by the Ministry Representative.

End dumping on the roadway followed by spreading will not be permitted.

Depositing and spreading the material on the roadway shall progress continuously without breaks, except as otherwise directed by the Ministry Representative.

At any break, including the end of each day's full width construction, a straight transverse construction joint shall be formed by cutting back into the completed work to form a true vertical face and by installing a temporary wooden header, if required by the Ministry Representative.

586.32.03 Compaction – Immediately following spreading, the material shall be compacted to a minimum 100% of the Standard Proctor Maximum Dry Density obtained by ASTM D698.

586.32.04 Compaction Equipment – The Contractor shall supply compacting equipment having sufficient capacity to produce the specified degree of compaction within the time limits specified. The use of compacting equipment which marks, ruts, cracks or otherwise loosens or damages the surface of the cement treated base will not be permitted.

Wherever compaction planes are formed in the cement treated base during spreading, shaping or compaction, they shall be removed by scarifying or tilling the surface to the required depth to completely remove the compaction plane and in such a manner as to thoroughly break up the compaction planes. Scarifying and pulverization may be performed by any suitable equipment – nail drag, weeder, disk harrow or rotary tiller that will satisfactorily perform the work required. The scarified material will be regraded and recompacted, to the satisfaction of the Ministry Representative.

586.32.05 Water – Water shall be incidental to the Work and shall be applied as required.

The surface of the uncompacted, partially compacted, and completely compacted material shall always be kept moist. An adequate water supply shall be maintained at the point of compacting.

586.32.06 Speed – The speed of operations shall be such that not more than forty-five minutes shall elapse between the time of starting mixing and the time of starting compaction. Shaping and final compaction shall be completed within two hours of commencement of mixing at the plant.

586.32.07 Finished Surface – The finished surface of the compacted base shall at all points be accurate to ± 15 mm of the grade and shall not, at any point, deviate by more than 10 mm from the bottom of a 3 m straight edge laid in any direction on the surface on either side of the crown of the roadway.

586.32.08 Temperature – No construction shall be undertaken when the temperature is below 4.0°C or when the conditions indicate that the temperature will fall below 4.0°C in the next 24 hours or if the subgrade is frozen.

586.33 Protection and Cover

586.33.01 General – Protection and cover shall be provided by one of the methods, A or B, given below and as directed in the Special Provisions.

(a) Method A: Crushed Gravel Course – A 25 mm crushed gravel shall be laid not more than 24 hours after the treated base is laid. The surface of the treated base and the overlying gravel shall be kept moist by the application of water as directed by the Ministry Representative.

(b) Method B: Bituminous Seal – As soon as possible after the cement treated base course has been finished, in accordance with preceding subsections, a bituminous seal shall be applied to the surface and sanded to protect the cement treated base course from drying out during the seven day curing period. The finished base course shall be kept continuously moist until the seal has been applied.

The bituminous sealing material shall be applied at a rate between 0.7 L/m^2 to 1.0 L/m^2 or as directed by the Ministry Representative in order to give complete surface coverage without excessive runoff.

Generally, this work shall be governed by the applicable provisions of SS 502.21. At the time of sealing, the surface of the cement treated base must be dense, free from all loose and extraneous material and shall contain sufficient moisture to prevent penetration of the bituminous material. If needed, water shall be

applied in sufficient quantity to fill the surface voids of the soil cement immediately before the bituminous material is applied. The seal cover shall be effectively maintained by the Contractor during the seven day curing period.

586.33.02 Protection – Any finished portion of the base course adjacent to construction, which is travelled by equipment used in constructing an adjoining section, shall be protected in such a manner as to prevent equipment from marring or damaging the completed work.

586.33.03 Temperature – At any time when the air temperature may be expected to fall to the freezing point, during the day or the night, sufficient above protection shall be given to the cement treated base to prevent its freezing for seven days after placement and until the soil cement has hardened.

586.34 Maintenance – The Contractor shall maintain the entire base course in good condition satisfactory to the Ministry Representative from commencement of construction until it has been paved with asphalt concrete. Maintenance shall include immediate repairs to any defects that may occur and shall be repeated at the Contractor's expense as often as may be necessary to keep the section continuously intact.

MEASUREMENT

586.81 Cement Treated Base – Cement treated base in place will be measured by the TONNE for the total mass actually accepted and incorporated in the base of the road.

PAYMENT

586.91 Cement Treated Base – Payment for CEMENT TREATED BASE will be at the Contract Unit Price per tonne for the total mass actually accepted and incorporated in the base of the road. Payment shall provide full compensation for supply or production of aggregate; supply, handling, hauling and storage of bulk cement and filler; provision of mixing water and plant mixing of cement treated base; haulage, spreading, watering, grading and compaction of the cement-treated base material in place; and for all overhead and all other incidentals necessary to complete the work in accordance with the specifications.